

Closing the structural gap $q \in \{3, 4\}$ for $(4, 3, 1^q)$: $\dim B = 5$ structurally for all $q \geq 3$

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Abstract

The May-13 afternoon paper `2026-05-13-4-3-1n-lower-bound-closed.tex` proved $\dim B_{j_0}^{(\lambda)} V_\lambda = 5 = f^{(3,2)}$ structurally for $\lambda = (4, 3, 1^q)$ at $q \geq 5$, with computational verification at $q \in \{3, 4\}$. The structural binding was the dim input on the sub-shape $\mu_A = (3, 3, 1^{q-1})$, whose own structural range required $q - 1 \geq 4$ (dim2-331 Theorem 1.3, propagated from the May-12 evening dim on $(3, 2, 1^{q-2})$).

The very-late prove session (`2026-05-13-very-late-low-q-dim.tex`) closed the $(3, 2, 1^{q'})$ dim formula structurally for *every* $q' \geq 1$ (Theorem A) and the $(4, 2, 1^{q'})$ dim formula for every $q' \geq 2$ (Theorem B). Substituting Theorem A as the input to dim2-331 extends the latter down to $q' \geq 2$ for $(3, 3, 1^{q'})$ (Lemma 8 below), unlocking the structural μ_A input on $(4, 3, 1^q)$ at $q - 1 \geq 2$.

Theorem. For $\lambda = (4, 3, 1^q)$ at every $q \geq 3$,

$$\dim B_{j_0}^{(\lambda)} V_\lambda = 5 = f^{(3,2)},$$

structurally. Combined with the May-13 afternoon paper for $q \geq 5$ and the computational match at $q \in \{3, 4\}$, this gives the closed-form dim formula $\dim B = f^{\lambda^{(\geq 2)}}$ for the entire $(4, 3, 1^*)$ family at $q \geq 3$.

The proof has *no new technical content*: every ingredient is already established in the May-13 papers (afternoon-2 dim2-331, afternoon-3 lower-bound-closed, very-late). The contribution is to identify that the new $(3, 2, 1^*)$ low- q closure (Theorem A) is exactly the input that propagates dim2-331 down to $q' \geq 2$, which in turn propagates the $(4, 3, 1^*)$ proof down to $q \geq 3$. This is a single propagation step along the existing SRD ladder.

Mod- P verification at $P = 100\,003$, $Q = 11$ confirms every intermediate dim value.

1 Setup and inputs

Notation as in the May-13 afternoon papers. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ obeying $(T_i - q)(T_i + 1) = 0$ and the standard braid relations. V_λ is the seminormal cell module for $\lambda \vdash n$ in asymmetric Murphy normalisation; $E_j^\pm$ are the q - and (-1) -eigenspaces of T_j . For $1 \leq k \leq n$,

$$R'_k := (T_{k-1} + 1) \cdots (T_1 + 1), \quad B_j^{(\lambda)} V_\lambda := R'_j \cdots R'_n V_\lambda, \quad j_0(\lambda) := \ell(\lambda) + 1.$$

We cite the following inputs verbatim from the May-13 papers.

Theorem 1 (A of very-late, 2026-05-13-very-late-low-q-dim.tex Theorem 1). For $\nu = (3, 2, 1^p)$ with $p \geq 1$ and $|\nu| = p + 5$,

$$\dim B_{j_0(\nu)}^{(\nu)} V_\nu = 2 = f^{(2,1)},$$

structurally.

Theorem 2 (B of very-late, same paper, Theorem 2). For $\nu = (4, 2, 1^p)$ with $p \geq 2$ and $|\nu| = p + 6$,

$$\dim B_{j_0(\nu)}^{(\nu)} V_\nu = 3 = f^{(3,1)},$$

structurally.

Theorem 3 (SRD-331 Corollary 4.3 + Lemma 3.2). For $\mu = (3, 3, 1^{q'})$ with $q' \geq 2$, $n_\mu := q' + 6$, $\ell_\mu := q' + 2$, $j_0(\mu) := \ell_\mu + 1$, and $\mu_1 := \mu \setminus \{c_1, c_*\} = (3, 2, 1^{q'-1})$ ($c_1 = (2, 3)$, $c_* = (q' + 2, 1)$), axial distance $|d| = q' + 2 \geq 4$,

$$B_{j_0(\mu)}^{(\mu)} V_\mu = (T_{n_\mu-4}+1)(T_{n_\mu-3}+1)(T_{n_\mu-2}+1) \Phi_*(B_{j_0(\mu_1)}^{(\mu_1)} V_{\mu_1}), \quad (1)$$

where $\Phi_* : V_{\mu_1} \hookrightarrow V_\mu$ is the $\{c_1, c_*\}$ -branching injection: $\Phi_*(v_{T'})$ is the $T_{n_\mu-1}$ - $(+q)$ -eigenvector of the 2-block at $\{c_1, c_*\}$. Φ_* is $H_q(S_{n_\mu-2})$ -equivariant and $\Phi_*(V_{\mu_1}) \subseteq E_{n_\mu-1}^+|_{V_\mu}$.

Proposition 4 (Adjacent-injectivity, May-19 + dim2-331 Theorem 3.4). For any $H_q(S_n)$ -module V and any $1 \leq a \leq b \leq n - 2$, the composition

$$(T_a+1)(T_{a+1}+1) \cdots (T_b+1)|_{E_{b+1}^+(V)} : E_{b+1}^+(V) \longrightarrow E_a^+(V)$$

is injective. Each factor $(T_j+1) : E_{j+1}^+(V) \rightarrow E_j^+(V)$ is well-defined (Hecke quadratic relation) and injective (May-19 adjacent disjointness $E_j^- \cap E_{j+1}^+ = 0$).

Theorem 5 (Four-branch reduction, May-13-UB Theorem 3.4 + Lemmas 3.5, 3.6). For $\lambda = (4, 3, 1^q)$ at every $q \geq 3$,

$$B_{j_0}^{(\lambda)} V_\lambda = \mathcal{O}_\lambda \left[\Phi_A^\lambda(B_{j_0(\mu_A)}^{(\mu_A)} V_{\mu_A}) + \Phi_B^\lambda(B_{j_0(\mu_B)}^{(\mu_B)} V_{\mu_B}) \right], \quad (2)$$

where $\mu_A = (3, 3, 1^{q-1})$, $\mu_B = (4, 2, 1^{q-1})$, $\mathcal{O}_\lambda = (T_{q+2}+1)(T_{q+3}+1)(T_{q+4}+1)(T_{q+5}+1)$ (four factors at indices $\ell, \ell+1, n-3, n-2$). **The reduction is fully structural at $q \geq 3$:** the third (μ_C -piece) and fourth (same-row piece) summands of the four-branch decomposition vanish at $q \geq 3$ (May-13-UB Lemma 3.5 + Lemma 3.6), citing the hook j -formulas on $\mu_C = (3, 2, 1^q)$ at $q \geq 1$ (Shift Lemma + hook j -formula chain) and on $\nu_2 = (4, 1^{q+1})$ at $q+1 \geq 2$ (Shift Lemma).

Remark 6. The reduction (2) does not depend on the j -formulas for the surviving sub-shapes μ_A, μ_B . It depends only on the j -formulas for the vanishing pieces μ_C and ν_2 , both of which are hooks and structural for all $q \geq 1$. This is the key observation enabling the structural upper bound at $q \geq 3$: the surviving pieces enter only via their *dimensions*, which my new inputs (Lemma 8 and Theorem 2) supply at $q \geq 3$.

Theorem 7 (Disjoint Φ -supports + image-in- E^+ , May-13 afternoon-3 Lemmas 2.1, 2.2). For $\lambda = (4, 3, 1^q)$ and the branching injections $\Phi_A^\lambda, \Phi_B^\lambda$ at corner pairs $\{c_1, c_3\}, \{c_2, c_3\}$ (where $c_1 = (1, 4)$, $c_2 = (2, 3)$, $c_3 = (q+2, 1)$):

1. $\Phi_A^\lambda(V_{\mu_A}) \cap \Phi_B^\lambda(V_{\mu_B}) = \{0\}$ inside V_λ (their seminormal SYT-supports are disjoint because the unordered cell pairs $\{c_1, c_3\}, \{c_2, c_3\}$ differ).
 2. $\Phi_A^\lambda(V_{\mu_A}) \cup \Phi_B^\lambda(V_{\mu_B}) \subseteq E_{n-1}^+|_{V_\lambda}$.
- Each Φ_X^λ is itself injective on V_{μ_X} .

2 Extended dim2-331: $\dim B^{(3,3,1^{q'})} = 2$ structurally at $q' \geq 2$

Lemma 8 (Extended dim2-331). *For $\mu = (3, 3, 1^{q'})$ at every $q' \geq 2$,*

$$\dim B_{j_0(\mu)}^{(\mu)} V_\mu = 2,$$

structurally.

Proof. The dim2-331 paper (2026-05-13-dim2-331-structural.tex) proves $\dim B^{(\mu)} = 2$ structurally at $q' \geq 4$ via three ingredients:

- (i) SRD-331 reduction formula (1) (Theorem 3): unconditional at $q' \geq 2$.
- (ii) Non-degeneracy of the corner-pair 2-block: $|d| = q' + 2 \geq 4$ at $q' \geq 2$.
- (iii) Dim of the inner space: $\dim B_{j_0(\mu_1)}^{(\mu_1)} V_{\mu_1} = 2$ for $\mu_1 = (3, 2, 1^{q'-1})$.

The original dim2-331 used the May-12 evening dim formula on μ_1 , requiring $|\mu_1| \geq 8$, i.e., $q' \geq 4$. Substituting Theorem A (very-late, Theorem 1 above) for the inner dim: $\dim B^{(\mu_1)} = 2$ for $\mu_1 = (3, 2, 1^{q'-1})$ at $q' - 1 \geq 1$, i.e., *every* $q' \geq 2$.

With this strictly stronger inner-dim input, the dim2-331 argument runs unchanged:

1. Upper bound: $B^{(\mu)} = \mathcal{P}_\mu \Phi_*(B^{(\mu_1)})$ (Theorem 3), Φ_* injective (axial distance $|d| = q' + 2 \geq 4$), $\dim B^{(\mu_1)} = 2$ (Theorem A applied at $p = q' - 1 \geq 1$). Hence $\dim B^{(\mu)} \leq \dim \Phi_*(B^{(\mu_1)}) = 2$.
2. Lower bound: $\Phi_*(B^{(\mu_1)}) \subseteq E_{n_\mu-1}^+|_{V_\mu}$ is 2-dimensional. The outer chain $\mathcal{P}_\mu = (T_{n_\mu-4}+1)(T_{n_\mu-3}+1)(T_{n_\mu-2}+1)$ is the composition of three adjacent factors, indices $n_\mu - 4, n_\mu - 3, n_\mu - 2$ in increasing order. By Proposition 4 with $a = n_\mu - 4$, $b = n_\mu - 2$, $\mathcal{P}_\mu$ acts injectively on $E_{n_\mu-1}^+|_{V_\mu}$. Hence $\dim \mathcal{P}_\mu \Phi_*(B^{(\mu_1)}) = \dim \Phi_*(B^{(\mu_1)}) = 2$, which gives $\dim B^{(\mu)} \geq 2$.

Combining, $\dim B^{(\mu)} = 2$. □

Remark 9. Lemma 8 introduces no new technical machinery. The dim2-331 argument was always shape-agnostic in ingredients (i), (ii), and the adjacent-injectivity chain. The only shape-specific dependency was the inner-dim input on μ_1 . Improving that input automatically improves the conclusion's range.

3 Main theorem: $\dim B^{(4,3,1^q)} = 5$ structurally at $q \geq 3$

Theorem 10 (Main). *For $\lambda = (4, 3, 1^q)$ at every $q \geq 3$,*

$$\dim B_{j_0}^{(\lambda)} V_\lambda = 5 = f^{(3,2)},$$

structurally.

Proof. We assemble the May-13 afternoon-3 proof at the new (lower) threshold. The structural reduction (2) gives

$$B_{j_0}^{(\lambda)} V_\lambda = \mathcal{O}_\lambda(W), \quad W := \Phi_A^\lambda(B_{j_0(\mu_A)}^{(\mu_A)} V_{\mu_A}) + \Phi_B^\lambda(B_{j_0(\mu_B)}^{(\mu_B)} V_{\mu_B}).$$

We will establish $\dim W = 5$ (Step 1) and that $\mathcal{O}_\lambda$ acts injectively on a containing subspace (Steps 2–3); together these yield $\dim \mathcal{O}_\lambda W = 5$, hence $\dim B_{j_0}^{(\lambda)} V_\lambda = 5$.

Step 1: $\dim W = 5$.

- By Lemma 8 applied at $q' = q - 1 \geq 2$ (i.e., $q \geq 3$): $\dim B_{j_0(\mu_A)}^{(\mu_A)} V_{\mu_A} = 2$, structurally.
- By Theorem 2 applied at $p = q - 1 \geq 2$ (i.e., $q \geq 3$): $\dim B_{j_0(\mu_B)}^{(\mu_B)} V_{\mu_B} = 3$, structurally.
- By Theorem 7(1), the seminormal V_λ -supports of $\Phi_A^\lambda(V_{\mu_A})$ and $\Phi_B^\lambda(V_{\mu_B})$ are disjoint, hence the two images intersect trivially; their sum is a direct sum. Each Φ_X^λ is itself injective on V_{μ_X} (distinct seminormal basis vectors go to vectors with disjoint 2-element supports). Hence

$$\dim W = \dim \Phi_A^\lambda(B^{(\mu_A)}) + \dim \Phi_B^\lambda(B^{(\mu_B)}) = \dim B^{(\mu_A)} + \dim B^{(\mu_B)} = 2 + 3 = 5.$$

Step 2: $W \subseteq E_{n-1}^+|_{V_\lambda}$. By Theorem 7(2), both summands of W lie in $E_{n-1}^+|_{V_\lambda}$, hence so does W .

Step 3: $\mathcal{O}_\lambda$ is injective on W . The outer chain is the four-factor product $\mathcal{O}_\lambda = (T_{q+2+1})(T_{q+3+1})(T_{q+4+1})(T_{q+5+1})$ with indices $a = q + 2 = \ell$ and $b = q + 5 = n - 2$. Proposition 4 applies: $\mathcal{O}_\lambda$ acts injectively on all of $E_{n-1}^+|_{V_\lambda}$, hence on the subspace W .

Conclusion. By Steps 1 and 3:

$$\dim \mathcal{O}_\lambda W = \dim W = 5.$$

By the four-branch reduction (2), $B_{j_0}^{(\lambda)} V_\lambda = \mathcal{O}_\lambda W$. Hence

$$\dim B_{j_0}^{(\lambda)} V_\lambda = \dim \mathcal{O}_\lambda W = 5.$$

(The argument gives both bounds simultaneously: $\dim B = \dim \mathcal{O}_\lambda W \leq \dim W = 5$ for the upper bound, and $\dim B = \dim \mathcal{O}_\lambda W = \dim W = 5$ for the lower bound, by injectivity.) $\square$

Remark 11 (Range of validity, sub-shape audit). We audit each structural input against the regime $q \geq 3$:

Ingredient	Required	Threshold at $\lambda = (4, 3, 1^q)$
Phase A four-branch decomposition (May-13-UB Thm 3.4)	$q \geq 3$	$q \geq 3$
μ_C -piece vanishes (May-13-UB Lem 3.5, hook j -form on $(3, 2, 1^q)$)	$q \geq 1$	$q \geq 3$
Same-row vanishes (May-13-UB Lem 3.6, hook j -form on $(4, 1^{q+1})$)	$q \geq 1$	$q \geq 3$
$\dim B^{(\mu_A)}$ structural	$q - 1 \geq 2$	$q \geq 3$
(Lemma 8, via Theorem A on $\mu_1 = (3, 2, 1^{q-2})$)	$q - 2 \geq 1$	$q \geq 3$
$\dim B^{(\mu_B)}$ structural (Theorem 2)	$q - 1 \geq 2$	$q \geq 3$
Disjoint Φ -supports (Theorem 7(1))	shape-agnostic	all q
$\Phi_X \subseteq E_{n-1}^+$ (Theorem 7(2))	$ d_X \geq 2$	all $q \geq 1$
Adjacent-injectivity of $\mathcal{O}_\lambda$ (Prop 4)	shape-agnostic	all q

Every input is available at $q \geq 3$; the binding constraint is the μ_A input via Lemma 8, which itself binds at $q' = q - 1 \geq 2$ via Theorem A at $p = q' - 1 = q - 2 \geq 1$.

Remark 12 (Comparison to the May-13 afternoon-3 paper). The May-13 afternoon-3 paper used the original dim2-331 (binding $q' \geq 4$) for the μ_A input, forcing $q - 1 \geq 4$, i.e., $q \geq 5$ as the binding threshold. Replacing the inner input with Theorem A (which sharpens dim2-331's μ_1 -input from $q' - 1 \geq 4$ to $q' - 1 \geq 1$) drops every subsequent threshold by 3 units: dim2-331 goes from $q' \geq 4$ to $q' \geq 2$, hence $(4, 3, 1^q)$ goes from $q \geq 5$ to $q \geq 3$.

4 Computational verification

Mod- P computation at $P = 100\,003$, $Q = 11$ via the `compute_dim_B.py` script (`scratch/2026-05-13-multi-corner`)

shape	n	$\dim V$	$\dim B$
$(4, 3, 1^3)$	10	525	5
$(4, 3, 1^4)$	11	1100	5
$(3, 3, 1^2)$	8	56	2
$(3, 3, 1^3)$	9	120	2
$(4, 2, 1^2)$	8	90	3
$(4, 2, 1^3)$	9	189	3
$(3, 2, 1^1)$	6	16	2
$(3, 2, 1^2)$	7	35	2

The first two rows verify the main theorem at the two new structural cases $q \in \{3, 4\}$. Rows 3–4 verify the $\mu_A = (3, 3, 1^{q-1})$ input at $q' \in \{2, 3\}$, matching Lemma 8. Rows 5–6 verify the $\mu_B = (4, 2, 1^{q-1})$ input at $p \in \{2, 3\}$, matching Theorem 2. Rows 7–8 verify the $\mu_1 = (3, 2, 1^{q-2})$ input for Lemma 8 at $p \in \{1, 2\}$, matching Theorem 1.

5 Outlook and the SRD-ladder shape audit

Remark 13 (Status of the $r = 2$ same-row club). After this paper, the structural j -formula club for $r = 2$ length-preserving-corner shapes covers the following stable regimes:

family	structural range	source
$(3, 2, 1^q)$	$q \geq 1$	very-late Theorem A
$(4, 2, 1^q)$	$q \geq 2$	very-late Theorem B
$(3, 3, 1^q)$	$q \geq 2$	Lemma 8 (this paper)
$(4, 3, 1^q)$	$q \geq 3$	Theorem 10 (this paper)
$(5, 2, 1^q)$	$q \geq 5$	afternoon-3 (only)

The next concrete target is $(5, 2, 1^q)$ at $q \in \{3, 4\}$. Following the same propagation analysis: the SRD-ladder template (see `srd-ladder-climb` memory) for $(5, 2, 1^q)$ has $\mu_A = (4, 2, 1^{q-1})$, $\mu_B = (5, 1^q)$ (hook), $\mu_C = (4, 1^{q+1})$ (hook), $\nu_1 = (3, 2, 1^q)$. With very-late Theorem B closing μ_A at $q - 1 \geq 2$ and Theorem A closing ν_1 at $q \geq 1$, the binding constraint should drop to $q \geq 3$. Pending: the actual same-row vanishing and reduction at $q \in \{3, 4\}$ inside the $(5, 2)$ -specific upper-bound paper.

Remark 14 (The matching set $\{q : \dim B = f^{\lambda(\geq 2)}\}$). Combined with the dim2-331-non-monotonicity remark in the decoration-iso-failed paper, this paper sharpens the picture of the matching set for $(4, 3, 1^q)$:

$$\{q \geq 0 : \dim B^{(4,3,1^q)} = f^{(3,2)} = 5\} = \{3, 4, 5, 6, \dots\} \stackrel{?}{=} \{q \geq 3\}.$$

At $q \in \{0, 1, 2\}$, direct computation (decoration-iso-failed Section 5) shows the failure values $\dim = 4, 7, 12$. The threshold $q \geq 3$ matches the conjecture $q_0^{\dim}(\hat{\lambda}) = \max(0, |\hat{\lambda}| - 2\ell(\hat{\lambda}))$ at $\hat{\lambda} = (4, 3)$: $|\hat{\lambda}| - 2\ell = 7 - 4 = 3$.